MSE4931 Ferroalloys

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Monday 15:00-16:50

- **Lecture 2:** Fabrication Methods of Ferroalloys-1

INTRODUCTION

Core Principle: The fundamental method in ferroalloy production is the reduction of metal oxides.

Metal Oxide Reduction

Core stoichiometric pathways in ferroalloy production

Carbothermic

Large-scale generic alloys.

$$SiO_2 + 2C \rightarrow Si + 2CO$$

$$MnO + C \rightarrow Mn + CO$$

Aluminothermic

Low-carbon requirements.

$$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$$

$$3MnO + 2Al \rightarrow 3Mn + Al_2O_3$$

Silicothermic

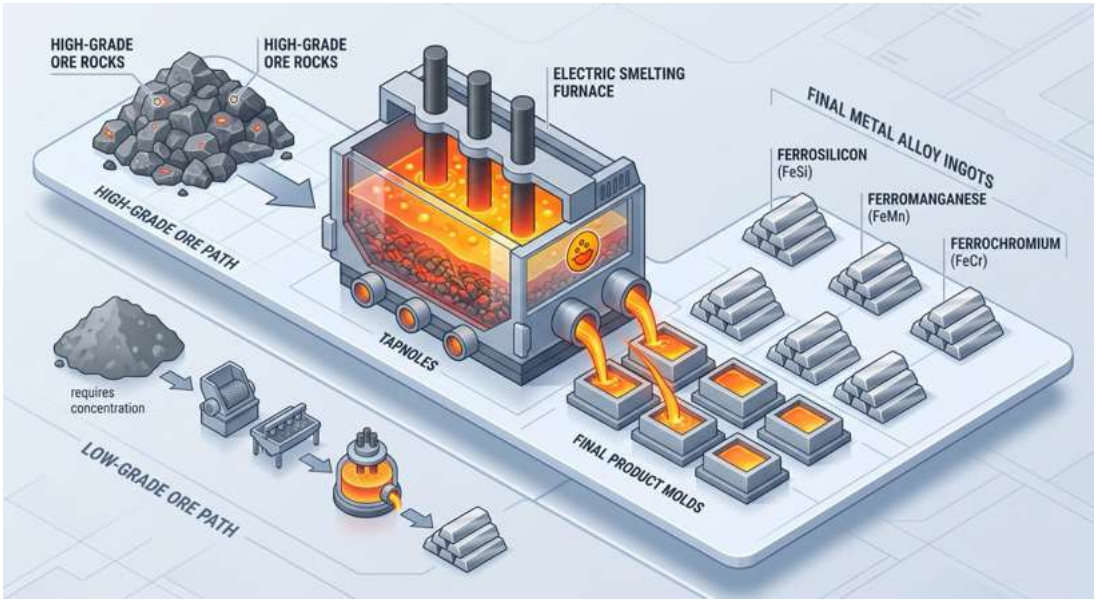
Highly exothermic processes.

$$2MoO_3 + 3Si \rightarrow 2Mo + 3SiO_2$$

INTRODUCTION

Optimization: Choosing the right reducing agent and operating conditions is critical to maximize the recovery of the valuable element. Raw Materials are ores or ore concentrates.

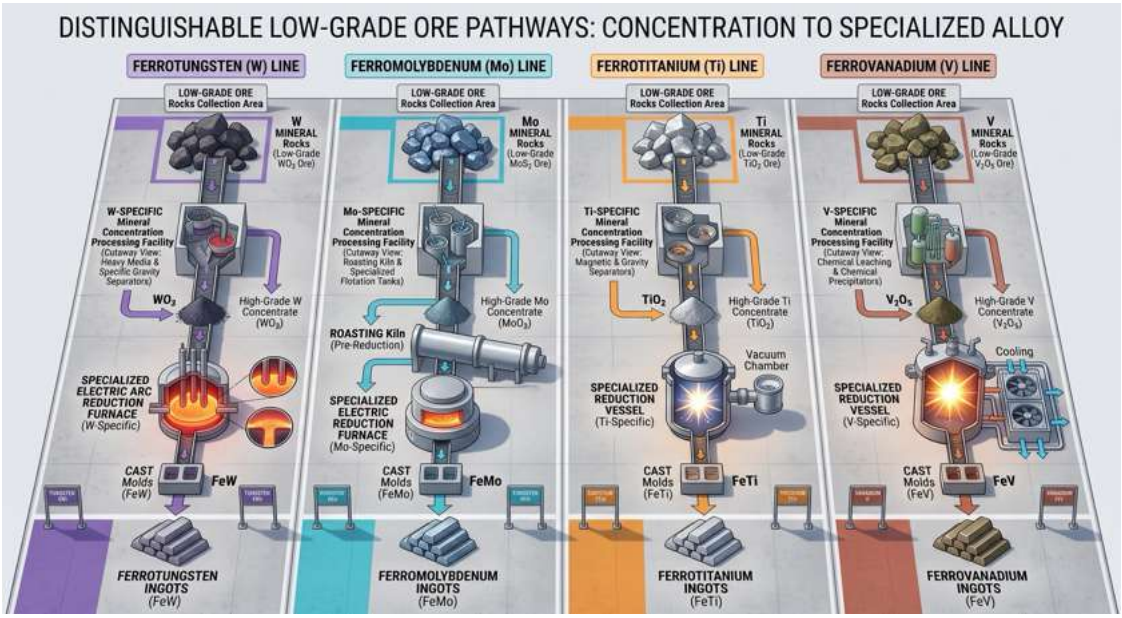
- **High-Grade Ores:** Used directly
  - Ferrosilicon, Ferromanganese, and Ferrochromium.



INTRODUCTION

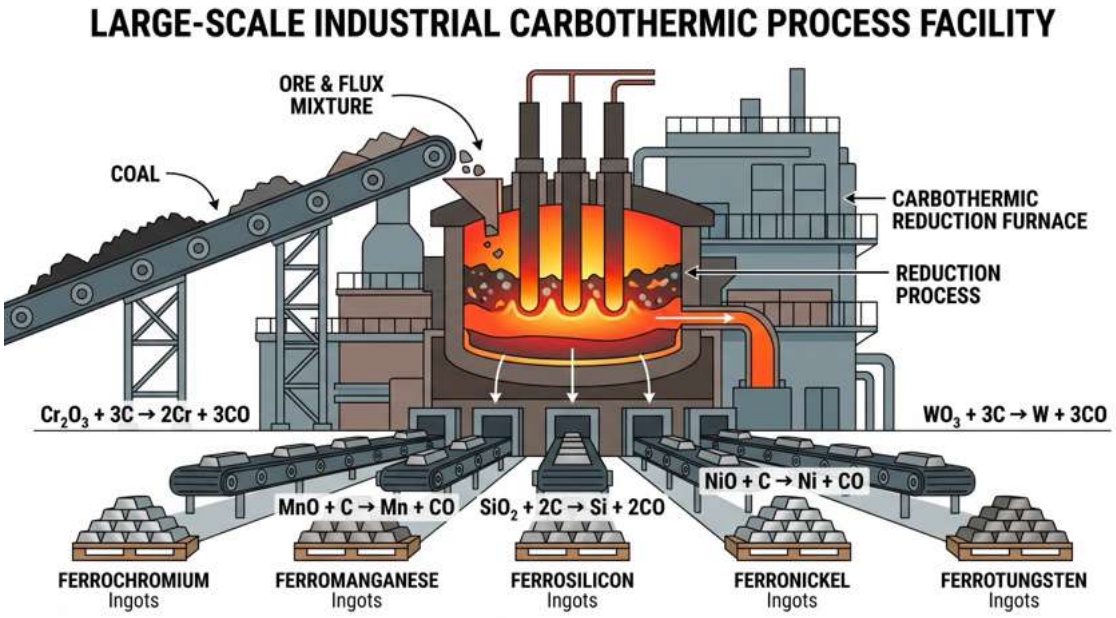
Optimization: Choosing the right reducing agent and operating conditions is critical to maximize the recovery of the valuable element. Raw Materials are ores or ore concentrates.

- **Low-Grade Ores:** Require concentration
  - Ferrotungsten, Ferromolybdenum, Ferrotitanium, and Ferrovanadium.



INTRODUCTION

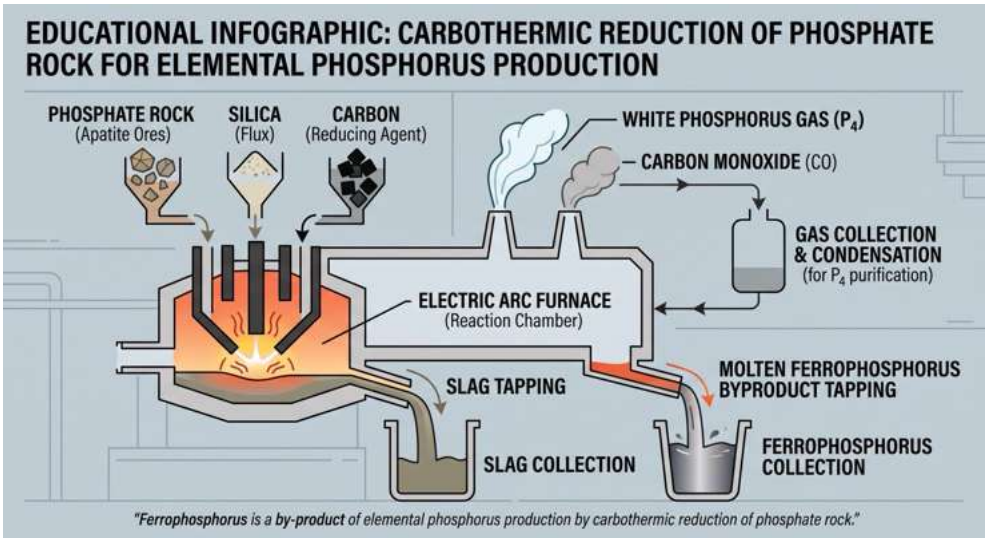
- **Carbothermic processes** are mainly used for large-scale production of ferrochromium, ferromanganese, ferrosilicon, ferronickel and ferrotungsten.



In actual industrial practice, **the iron oxides** present in the ores are also simultaneously reduced to form the "ferro" base of the alloy, but these formulas represent the primary reduction of the target metal oxides.

INTRODUCTION

Ferrophosphorus is a by-product of elemental phosphorus production by the carbothermic reduction of phosphate rock.



1. Main Elemental Phosphorus Reaction

$$2Ca_3(PO_4)_2 + 6SiO_2 + 10C \rightarrow P_4 + 10CO + 6CaSiO_3$$

2. Ferrophosphorus By-Product Reactions

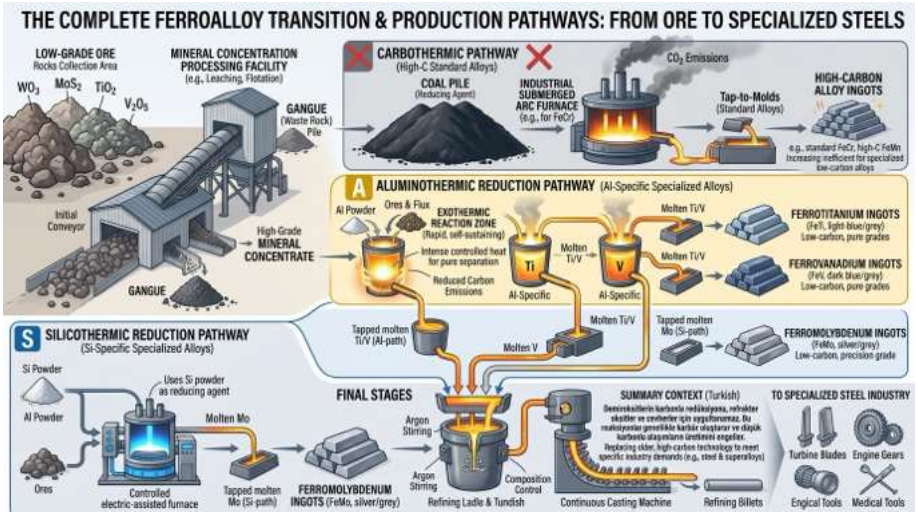
$$Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$$

$$2Fe + P \rightarrow Fe_2P$$

INTRODUCTION

For ferroboron, ferrotitanium, ferrovanadium and ferromolybdenum production, **metallothermic processes**, mainly aluminothermic and silicothermic reduction, have largely replaced carbothermic processes.

The Problem: Carbothermic reduction uses **carbon/coal.** While great for standard alloys, for reactive metals (Ti, V, B, Mo), carbon causes unwanted carbide formation, resulting in a high-carbon product.



The Solution (Right): Metallothermic processes (Aluminothermic using Al, Silicothermic using Si) provide highly exothermic, rapid reactions that bypass carbon entirely, resulting in the high-purity, low-carbon alloys required by modern steelmaking.

Overcoming Carbothermic Limits with Metallothermic Reduction

- 1. The Limitations of Carbon Reduction
  - Conventional carbon reduction is unsuitable for refractory oxides and ores.
  - Products generally require low carbon contents; strong carbide formation must be avoided.
  - Conventional refractories cannot withstand prolonged reduction at high temperatures.
  - Example Unwanted Reaction:
  - $TiO^2 + 3C \rightarrow TiC + 2CO$

- 2. The Metallothermic Solution
  - Metallothermic reactions using reactive metals (especially Al) are preferred.
  - Rapid Kinetics: Provides rapid reaction rates, avoiding prolonged heating.
  - Self-Sustaining: Highly exothermic reactions generate high temperatures internally.
  - Example Solution Reaction:

$$3TiO_2 + 4Al \rightarrow 3Ti \text{ (Pure, Low–Carbon Metal)} + 2Al_2O_3 + \text{Massive Heat}$$

Overcoming Carbothermic Limits with Metallothermic Reduction

Specific examples of refractory oxides and ores that are typically reduced using metallothermic processes (like aluminothermic or silicothermic reduction) because conventional carbon reduction would lead to strong, unwanted carbide formation and require unsustainably high temperatures:

Titanium Dioxide (TiO_2): Found in rutile or ilmenite ores. If reduced with C, it forms titanium carbide (TiC) instead of pure Ti.

Vanadium Pentoxide (V_2O_5): Extracted from vanadiferous ores. It requires metallothermic reduction to produce low-C ferrovanadium.

Niobium Pentoxide (Nb_2O_5): Found in columbite ores. Nb is a strong carbide former, so aluminum is used to produce pure ferroniobium.

Tungsten Trioxide (WO_3): Found in wolframite or scheelite ores. Metallothermic reduction is preferred to produce high-purity, low-C ferrotungsten.

Molybdenum Trioxide (MoO_3): Derived from molybdenite. It is typically reduced using Si or Al to avoid the carbon contamination that would ruin the resulting ferromolybdenum.

Boron Trioxide (B_2O_3): Found in borate minerals. It is reduced with Al to produce ferroboron, bypassing the carbon pathway entirely.

Zirconium Dioxide (ZrO_2): Found in zircon (ZrO_2). Zr is highly reactive with C, making metallothermic reduction essential for producing low-C Zr alloys.

For all of these refractory oxides, using reactive metals like Al provides the rapid reaction rates and intense, self-sustaining heat needed to successfully reduce the metal while maintaining a low carbon content.

Metalothermic Reduction Reactions

Aluminothermic Reduction Reactions

Producing low-carbon specialized ferroalloys using highly exothermic aluminum reduction.



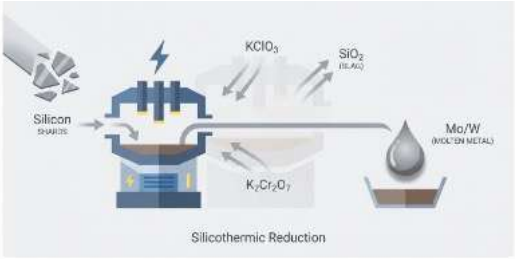
1. **Ferrovanadium (FeV)**
$$3V_2O_5 + 10Al \rightarrow 6V + 5Al_2O_3$$

2. **Ferrotitanium (FeTi)**
$$3TiO_2 + 4Al \rightarrow 3Ti + 2Al_2O_3$$

3. **Ferroboron (FeB)**
$$B_2O_3 + 2Al \rightarrow 2B + Al_2O_3$$

Silicothermic Reduction Reactions

Producing low-carbon refractory ferroalloys using silicon as the primary reducing agent.



1. **Ferromolybdenum (FeMo)**
$$2MoO_3 + 3Si \rightarrow 2Mo + 3SiO_2$$

2. **Ferrotungsten (FeW)**
$$2WO_3 + 3Si \rightarrow 2W + 3SiO_2$$

3. **Iron Oxide Base Reduction**
$$2FeO + Si \rightarrow 2Fe + SiO_2$$

The Evolution of Metallothermic Processes

1825

Jöns Jacob Berzelius

Early reduction of K2TaF7 using Sodium (Na).

1828

Friedrich Wöhler

Early reduction of AlCl3 using Sodium (Na).

1898

Hans Goldschmidt

Widely regarded as the father of aluminothermic processes.

Described the production of low-carbon metals with high melting points without the need for external heating. The charge consists of an exothermic mixture, where only a small portion is ignited to initiate the reaction.

Reduction Thermodynamics

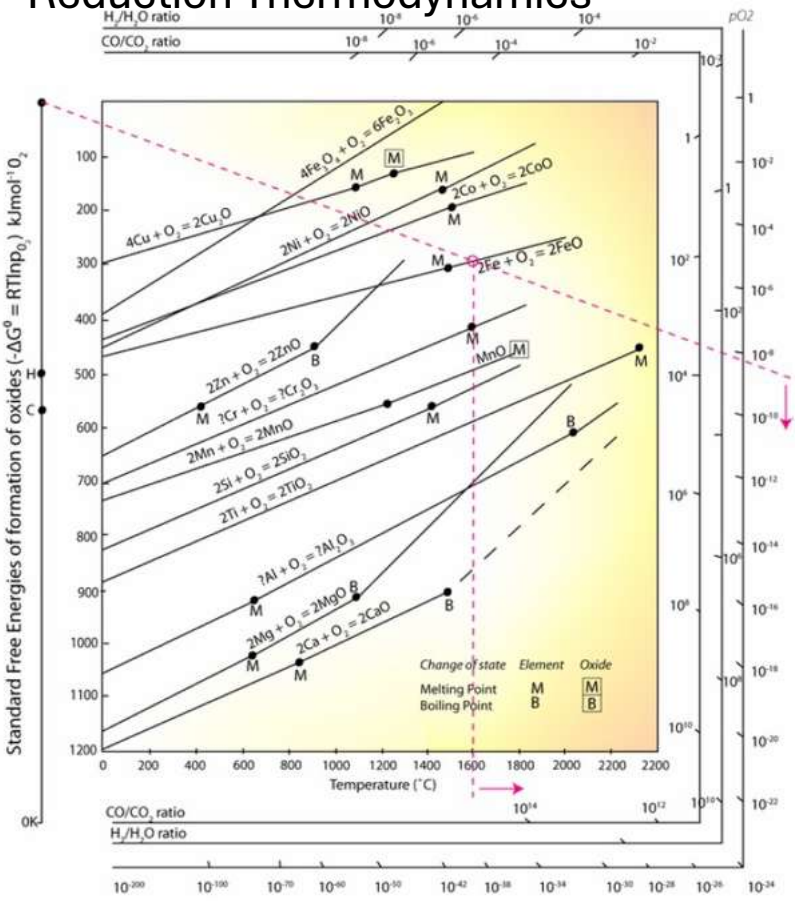
- The Gibbs free energy of formation provides a quantitative measure of oxide stability. The relationship between ΔG and the enthalpy of formation is:

$$\Delta G_T^o = \Delta H - T.\Delta S$$

- S denotes entropy, which is zero or positive and relates to increasing atomic disorder as the temperature rises.

- Oxides generally become less stable as temperature increases. At a given temperature, a metal whose oxide has a more negative free energy of formation can reduce an oxide with a less negative value.

Reduction Thermodynamics

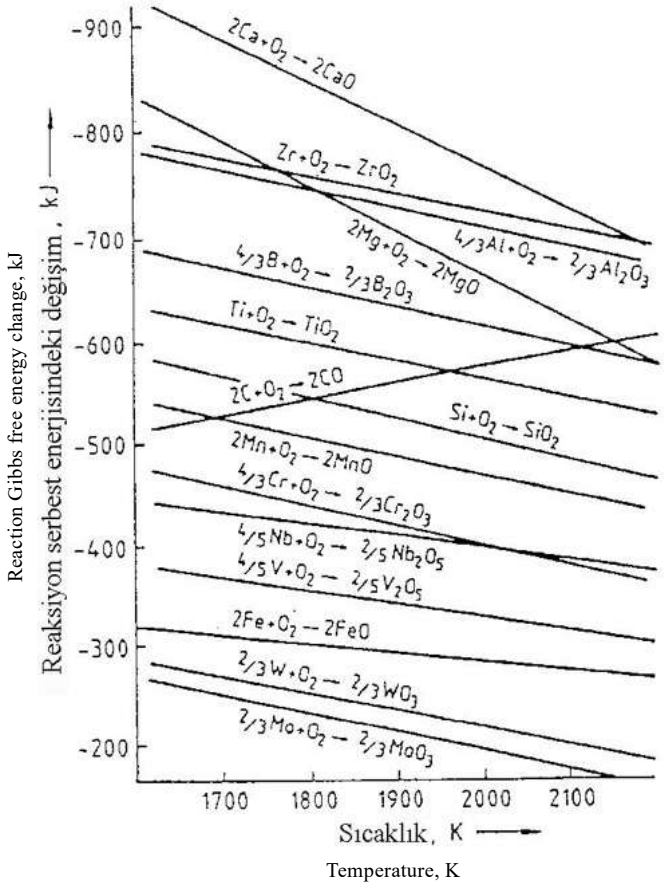


- Reduction reactions depend on the stability of oxides at high temperatures.
- Figure shows the variation of standard Gibbs free energies of oxide formation with temperature.
- The diagram allows comparison of oxide stability. For convenience, the free energy values are expressed per mole of oxygen.

Figure: °C and –ΔG°

More stable oxide is lower in the diagram

Reduction Thermodynamics



⦿ Gibbs Free Energy Changes for Oxide Formation

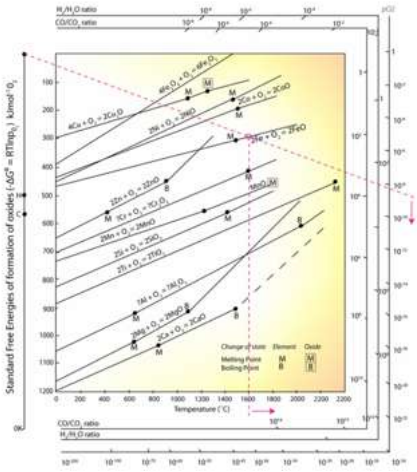
MoO_3 is the least stable oxide shown, whereas CaO is the most stable. Accordingly, MoO_3 is readily reduced, while CaO is very difficult to reduce.

Figure: K and ΔG°

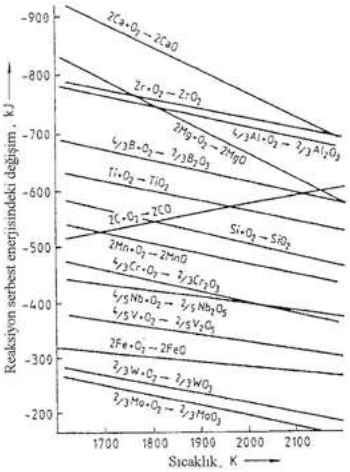
More stable oxide is higher in diagram

Reading and using Ellingham diagrams

Oxide stability and reduction at a chosen temperature



A: °C and −ΔG°



B: K and ΔG°

More stable oxide: lower in A, higher in B

Thermodynamic feasibility does not determine reaction rate.

1 Read the axes

Choose T, then read the oxide line. Compare ΔG° values per mole of O_2. More negative ΔG° means a more stable oxide.

2 Select a reducing agent

Its oxidation must have a more negative ΔG°.

ΔG°reduction = ΔG°reductant − ΔG°oxide < 0

Line intersections mark standard-state limits.

3 Read the pink example in A

At 1600 °C, locate the Fe/FeO line. Extend the ray from the 0 K origin through this point to the pO_2 scale: approximately 10^{-9}. Lower pO_2 favors Fe.

4 Apply diagram B at 1900 K

Al oxidation has a more negative ΔG° than Cr oxidation. Al can therefore reduce Cr_2O_3:

$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$

Reduction Thermodynamics

Gibbs free energy and oxide stability

For oxidation reactions compared at the same temperature and per mole of O_2

The thermodynamic relationship

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

ΔG°	Standard Gibbs free energy change
ΔH°	Standard enthalpy change
ΔS°	Standard entropy change
T	Absolute temperature in kelvin (K)

Worked example (illustrative values)

Assume

ΔH° = −500 kJ/mol O_2. and

ΔS° = −0.10 kJ/(mol O_2·K).

1000 K: ΔG° = −500 + 100 = −400 kJ/mol O_2

1500 K: ΔG° = −500 + 150 = −350 kJ/mol O_2

Reduction rule: a metal can reduce another oxide when forming its own oxide gives a more negative ΔG°.

What the equation tells us

More negative ΔG° means greater oxide stability relative to the metal and oxygen.

ΔS° can be positive, negative, or zero. It is the entropy change of the reaction:

$$\Delta S^\circ = \Sigma S^\circ_{products} - \Sigma S^\circ_{reactants}$$

Most metal oxidation reactions consume O_2 gas and form condensed oxides, so ΔS° is negative.

As T rises, ΔG° usually becomes less negative.

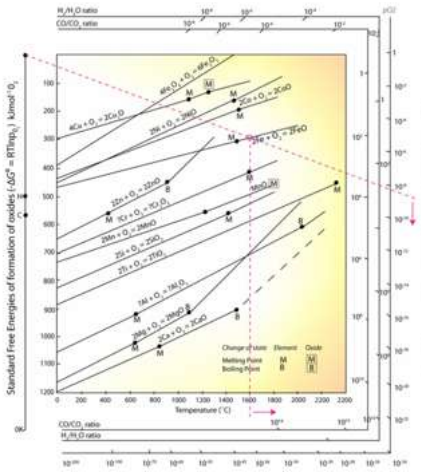
an a conventional ΔG°–T plot,

slope = −ΔS°.

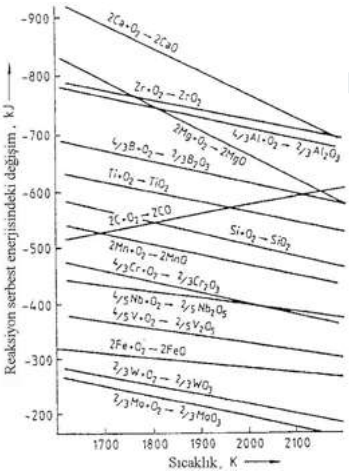
This explains the positive slope of most metal oxide formation lines.

Oxide stability: examples from Ellingham diagrams

$\Delta G° = \Delta H° - T\Delta S°$ More negative $\Delta G°$ means a more stable oxide. Compare at the same T and per mole of O_2.



A: °C and −ΔG°



B: K and ΔG°

More stable oxide: lower in A, higher in B

Example 1: Fe/FeO at 1600 °C (A)

1. Locate the Fe/FeO point on the pink vertical.
2. Follow the pink ray from the 0 K origin through this point to the outer pO_2 scale.
3. Read an equilibrium pO_2 of about 10^{-9} on the pressure scale shown.

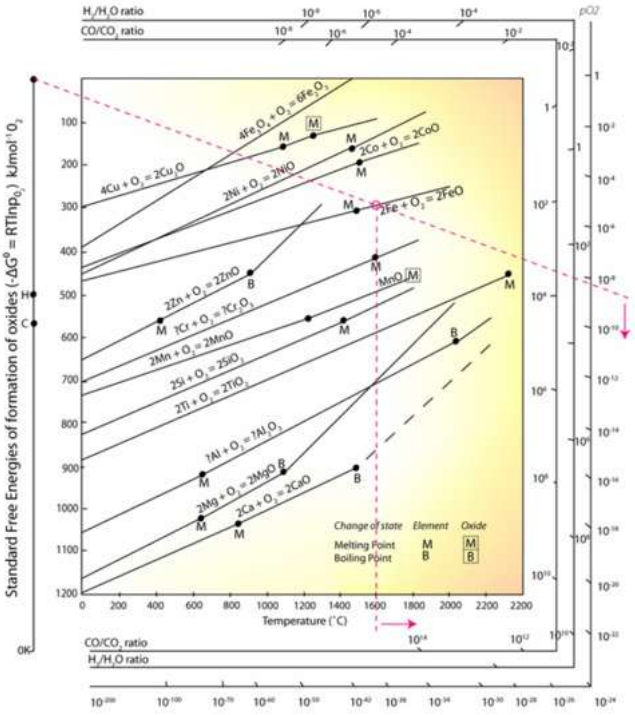
Below this equilibrium pressure, FeO reduction to Fe is thermodynamically favorable.

Example 2: Al reduces Cr_2O_3 at 1900 K (B)

Approximate oxidation values (kJ/mol O_2):
Al to Al_2O_3: −760
Cr to Cr_2O_3: −420
$\Delta G°_{reduction} \approx -760 - (-420) = -340$

$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$
$\Delta G° \approx 1.5 \times (-340) = -510$ kJ/mol Cr_2O_3
The negative value supports reduction by Al.

Reduction Thermodynamics



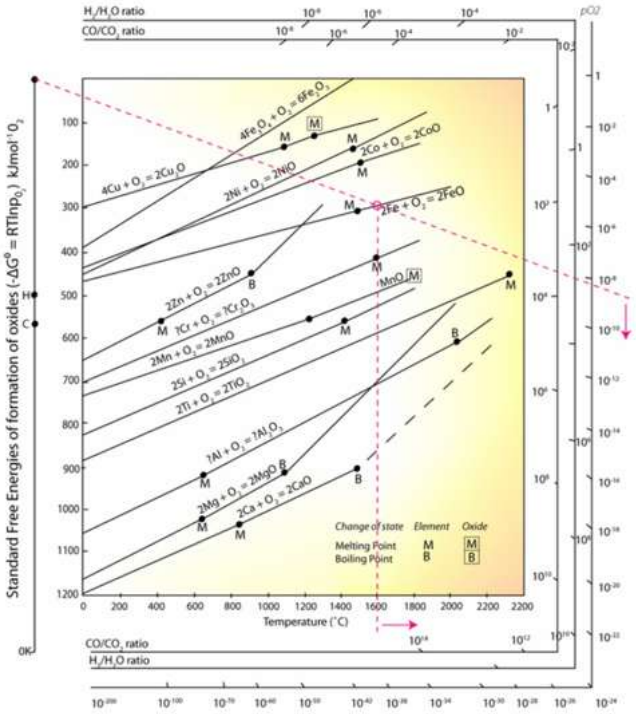
- The diagram also indicates which elements can serve as reducing agents. Under suitable conditions, an element that forms a more stable oxide can reduce the oxide of a metal that forms a less stable oxide.
- In practice, effective reduction requires a free energy difference of several tens of thousands of calories between the oxide being reduced and the oxide of the reducing agent.
- A small difference prevents complete reduction and increases the residual reducing-agent content of the alloy.

Reducing agents in Ellingham diagrams

At the same temperature, an element that forms a more stable oxide can reduce a less stable metal oxide under suitable conditions.

THERMODYNAMIC COMPARISON

ΔG°reduction = ΔG°oxidation of reductant − ΔG°oxidation of metal



Compare both oxidation reactions on the same O_2 basis. A negative result favors reduction.

Large negative ΔG°

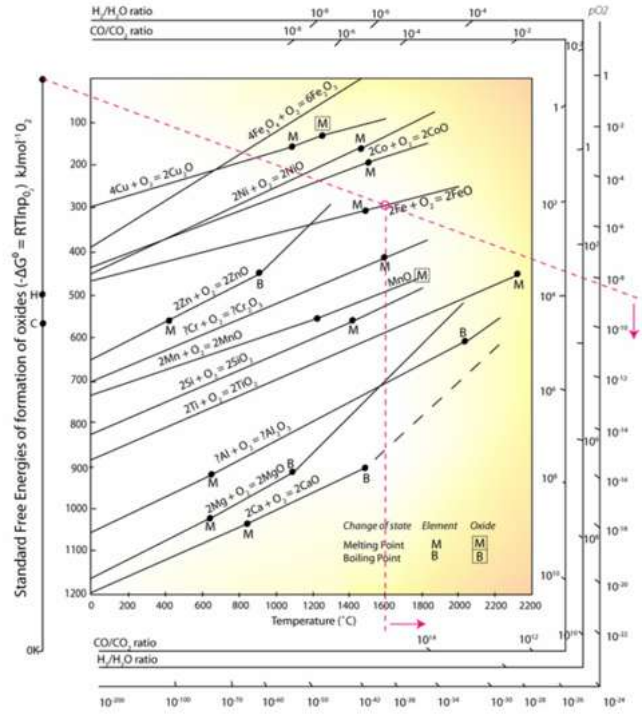
Stronger driving force favors more complete oxide reduction and lower equilibrium reductant residue.

Small negative ΔG°

Weaker driving force may leave unreduced oxide and require more reductant remaining in the alloy.

A gap of several tens of kcal per stated reaction basis may be needed. The required gap depends on temperature and activities. Thermodynamics does not give the reaction rate.

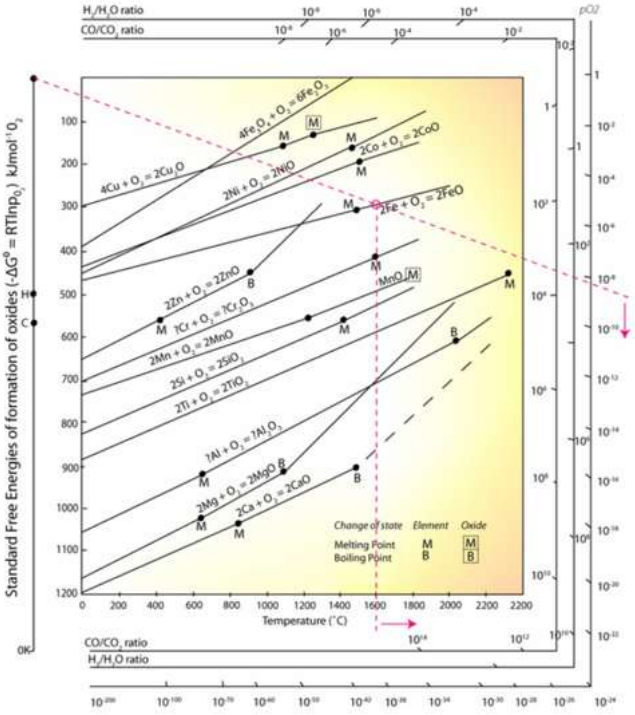
Reduction Thermodynamics



- Some elements form two or more oxides.
- Lower oxides are more stable than higher oxides.
- Therefore, reducing lower oxides to the elemental state is more difficult than reducing higher oxides.

Fe_2O_3 Fe_3O_4 FeO

Reduction Thermodynamics



- Iron or iron oxides facilitate reduction. Iron dissolves the reduced elements and forms alloys with them, lowering their activity and effectively removing them from the reaction zone.
- This also helps prevent reoxidation.

Reduction Thermodynamics

How iron promotes oxide reduction

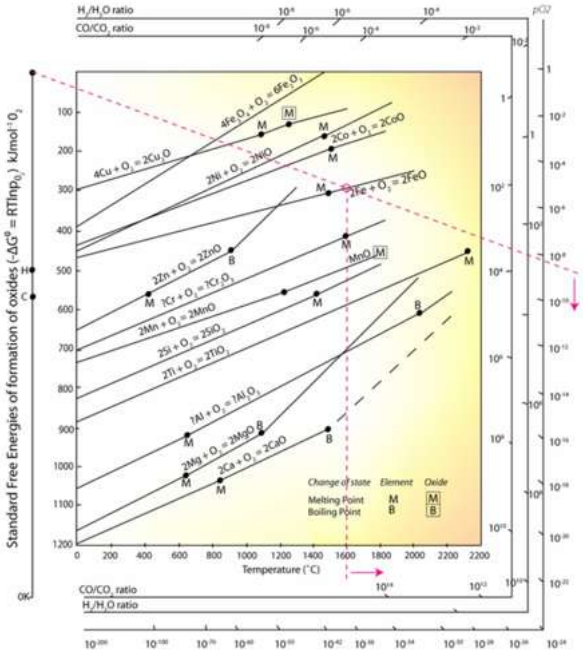
Metallic iron collects reduced elements in an alloy. Dissolution can lower their activity, increasing the thermodynamic driving force for reduction.

Why alloying helps

MO + C → M(in Fe) + CO

$$\Delta G = \Delta G^\circ + RT \ln[aM(pCO/p^\circ)/(aMO\ aC)]$$

At fixed temperature and other activities, lower aM makes ΔG more negative. The metal stays in the alloy with a lower chemical potential.



Ferrochromium

Reduced Cr dissolves in an Fe–Cr alloy.
Lower Cr activity favors further Cr_20_3 reduction.

Ferromanganese

Reduced Mn enters an Fe–Mn–C alloy.
Lower Mn activity can promote MnO reduction by carbon.

Iron oxides supply this alloy phase after reduction to Fe. Lower activity can limit reoxidation, but a reducing atmosphere and control of slag oxygen potential remain necessary.

Reduction Thermodynamics

Melting behavior and reductant selection

Alloy composition can lower the liquidus temperature

Many Fe-based compositions melt below the corresponding pure metal.
This can reduce the temperature needed to maintain a liquid metal bath.

Example:
the Fe–C eutectic (4.3 wt% C) melts at about 1147 °C, compared with about 1538 °C for pure iron (Fe–Fe_3C diagram).

Common reducing agents in ferroalloy production

Carbon	Usually the lowest-cost choice, but carbon pickup is common. Examples: high-carbon ferrochromium and ferromanganese.
Silicon	Supports low-carbon grades, but residual Si needs control. Example: silicothermic production of low-carbon ferrochromium.
Aluminium	Strong, often exothermic reductant with little direct carbon input. Example: aluminothermic ferrovanadium. Residual Al needs control.

Smelting must also satisfy oxide-reduction and slag-fluidity requirements.
A lower alloy liquidus alone does not guarantee a lower operating temperature.

Reduction Thermodynamics

Carbon reduction and carbide formation

Carbon is suitable when the product specification tolerates carbon

Coke, coal or charcoal provide an economical reductant.
Carbon can dissolve in the metal and form carbides, so the final alloy may need refining.

Industrial examples

High-carbon FeCr and FeMn

Carbon pickup and carbide formation are compatible with high-carbon grades.
Examples: Cr_7C_3 and Mn_7C_3.

Mo, V and Ti: carbide risk

Stable carbides can favor carbide-rich products rather than low-carbon alloys.
Examples: Mo_2C, VC and TiC.

Example: $TiO_2 + 3C \rightarrow TiC + 2CO$

Ferrosilicon is an important qualification

Carbon reduces silica, but high-Si FeSi has limited carbon solubility.
Using carbon as a reductant does not automatically produce a high-carbon alloy.

Silicon-rich alloys and carbon control

A silicon-rich alloy can help produce a lower-carbon product

In many Fe–Cr–Si and Fe–Mn–Si compositions, increasing silicon lowers carbon solubility and changes the stability of metal carbides.

Thermodynamic explanation

Metal–silicon interactions can lower the activity of the reduced metal. This favors oxide reduction into the alloy and can suppress metal-carbide formation.

Ferrochromium–silicon

A Si-rich Fe–Cr alloy can have less dissolved carbon than high-carbon FeCr and serve as a silicothermic reductant.

Silicomanganese

Increasing Si generally lowers carbon solubility in Fe–Mn–Si melts.
High-Si grades can supply low-carbon routes.

High Si does not guarantee low carbon: temperature, composition and carbon supply matter. SiC can still form. A low-carbon intermediate may require further treatment to lower Si.

Reduction Thermodynamics

Ferroalloy recovery and the Gibbs phase rule

Goal: maximize valuable-metal recovery while meeting alloy specifications

Recovery (%) = 100 × mass of element in product / mass of element in feed

Example: 90 kg Cr in ferrochromium from 100 kg Cr in the feed gives 90% recovery. Losses include unreduced oxides, metal droplets in slag, and dust or vapor.

Equilibrium constraints

$$f + p = c + 2$$

f: independent intensive variables
p: number of coexisting phases
c: independent components, accounting for reaction constraints
2: temperature and pressure

Example: binary Fe–Cr alloy

Liquid + solid: c = 2 and p = 2
f = 2 − 2 + 2 = 2
At fixed pressure: f = 1
Temperature sets equilibrium phase compositions in the two-phase region.

In smelting, metal, slag and gas can coexist as distinct phases.

Recovery depends on temperature, gas pressures, and metal/slag activities, plus kinetics and separation. The phase rule counts equilibrium freedoms. Recovery requires thermodynamic data and mass balances.

Reduction Thermodynamics

Phase rule at constant pressure

Fixed pressure removes one degree of freedom

f + p = c + 1

Idealized ternary system: c = 3 and p = 2

Assume two equilibrium liquid phases: **slag and alloy.**
Components: reduced metal, reductant and oxygen (e.g., Fe, Si, O).

f = 3 – 2 + 1 = 2

- Choose temperature and one independent composition variable in either phase.
- Equilibrium fixes the remaining phase-composition variables.

Example: idealized Fe–Si–O

At fixed pressure, specify temperature and Si content in the alloy, where they are valid independent variables. Equilibrium then constrains slag and alloy compositions.

Real furnaces usually contain additional components, gas and sometimes solids. Recount c and p for the actual system. Phase amounts require an overall mass balance.

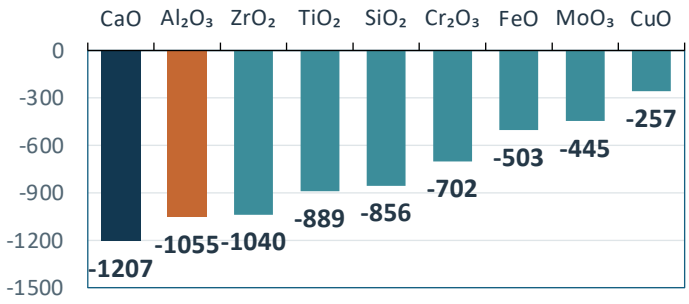
Principles of metallothermic reduction

A reducing metal removes oxygen by forming a more stable oxide.

Compare oxidation ΔG° at the same temperature and per mole of O_2.

Oxide stability at 298 K (25 °C)

Standard oxidation ΔG°, kJ per mol O_2
(rounded NIST–JANAF values)



More negative ΔG° means stronger oxygen affinity.

At 298 K, Al can thermodynamically reduce the seven oxides to its right. CaO is more stable than Al_2O_3.

Worked example: chromium production

$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$

At 298 K, per mole of oxide:
ΔG°f(Al_2O_3) ≈ −1582 kJ/mol
(1582/1.5: 1054.6 KJ)
ΔG°f(Cr_2O_3) ≈ −1053 kJ/mol
ΔG°reaction ≈ −1582 – (−1053)
≈ −529 kJ/mol Cr_2O_3

The negative value favors oxygen transfer from chromium oxide to aluminum.

At 1900 K, ΔG°reaction ≈ −425 kJ/mol: the same reduction remains favorable.

Principles of metallothermic reduction

Low-carbon ferroalloys for stainless steel

Silicon and aluminum reduce oxides while limiting carbon pickup.

Why carbon control matters

Ferroalloys supply Cr, Mn or Ti to the steel. Their carbon also enters the melt, so low-carbon additions help meet the final steel specification.

During welding or unsuitable heat exposure, carbon can form chromium-rich carbides at grain boundaries. This depletes nearby Cr and increases intergranular corrosion risk.

Lower carbon reduces this sensitization risk and helps preserve corrosion resistance.

Examples: low-carbon FeCr and FeMn. Low-carbon FeTi supplies Ti for stabilization without adding unnecessary C.

Stainless steelmakers can also refine high-carbon ferroalloys by AOD (argon-oxygen decarburization) and VOD (vacuum oxygen decarburization) to remove carbon.

Metallothermic routes

Silicothermic reduction: Si

$$2Cr_2O_3 + 3Si \rightarrow 4Cr + 3SiO_2$$

Often uses FeSi or a Si-rich alloy to produce low-carbon ferrochromium. SiO_2 enters the slag.

Aluminothermic reduction: Al

$$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$$

Used for low-carbon alloys, including FeTi. With an Fe-bearing charge, reduced metal joins the alloy and Al_2O_3 enters the slag.

Si and Al replace carbon as the reductant. Feedstock purity still controls final carbon.

Principles of metallothermic reduction

Heat balance and reductant selection

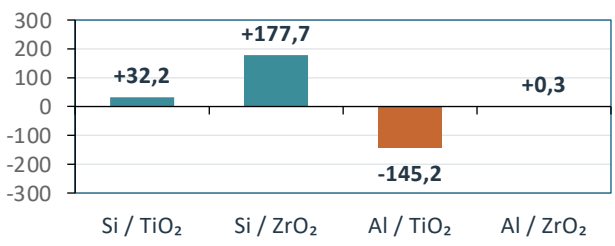
Enough heat sustains the process. Favorable ΔG determines which metal can form.

When reaction heat is insufficient

Preheat the charge or provide external heating. A supplementary exothermic reaction can also add heat. $KClO_3$ is a historical oxidizing booster that reacts with the reductant, rather than a fuel.

Reduction feasibility at 1000 K

ΔG°reaction, kJ per mol MO_2 (M = Ti or Zr)



ΔG° < 0 favors reduction in the standard-state comparison.

Calculated from NIST–JANAF data at 1000 K. Pure oxide phases and elemental standard states.

Why Si is unsuitable for low-Si Ti or Zr

TiO_2 and ZrO_2 are more stable than SiO_2. Their direct reduction by Si to pure metal has positive ΔG° at the temperature shown.

$$MO_2 + Si \rightarrow M + SiO_2$$

Reduction into a Si-rich alloy or silicide can become favorable because alloying lowers the activity of Ti or Zr. The product then contains substantial silicon.

Al offers a stronger alternative for TiO_2.

ZrO_2 reduction by Al remains conditional: check temperature, alloy and slag activities.

Principles of metallothermic reduction

Silicon can reduce Mn, Cr, W and Mo oxides under suitable process conditions.

Why **reduction** is possible

Balanced oxygen-transfer reactions

Manganese $2MnO + Si \rightarrow 2Mn + SiO_2$

Chromium $2Cr_2O_3 + 3Si \rightarrow 4Cr + 3SiO_2$

Tungsten $2WO_3 + 3Si \rightarrow 2W + 3SiO_2$

Molybdenum $2MoO_3 + 3Si \rightarrow 2Mo + 3SiO_2$

In ferroalloy production, the reduced metal collects in the Fe alloy and SiO_2 enters the slag.

Exothermic does not mean self-sustaining

- $\Delta G < 0$: reduction is thermodynamically favorable.
- $\Delta H < 0$: the reaction releases heat.
- Both must be assessed for the actual charge.

Many charges still need furnace heating

Reaction heat may be insufficient to heat and melt the alloy and slag, overcome heat losses, and maintain the temperature needed for reduction and metal–slag separation.

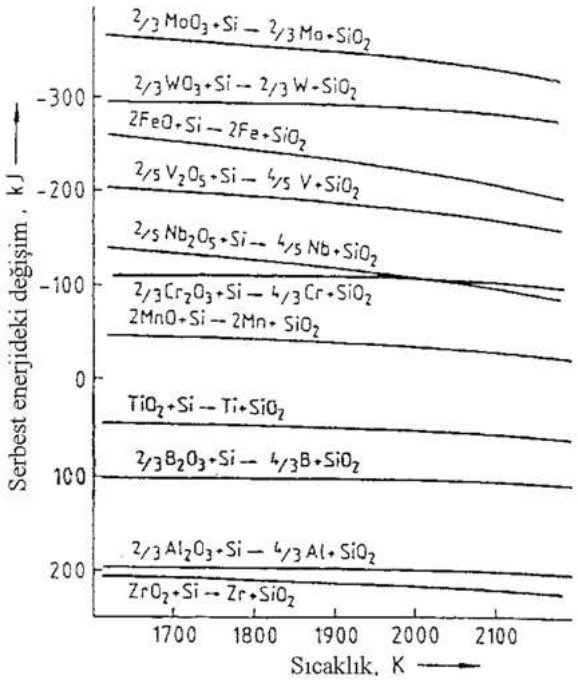
Important exception: MoO_3 reduction

$2MoO_3 + 3Si$: $\Delta H°_{298} \approx -1242$ kJ
This is about 621 kJ released per mol Mo.
A suitable FeMo charge can sustain itself after ignition without continuous furnace heat.

Principles of metallothermic reduction

Reading a silicothermic reduction diagram

Use ΔG to judge feasibility; use a separate heat balance to assess furnace heating.



1 Choose the temperature

At 1900 K (≈ 1627 °C), locate each reaction curve and read its ΔG from the vertical axis.

2 Check the sign — this axis is reversed

Above zero: $\Delta G < 0$ → favorable reduction.
Below zero: $\Delta G > 0$ → unfavorable in the reference states represented by the graph.

3 Compare approximate readings at 1900 K

MoO_3: −350 kJ **MnO: −35 kJ**

TiO_2: +50 kJ **ZrO_2: +220 kJ**

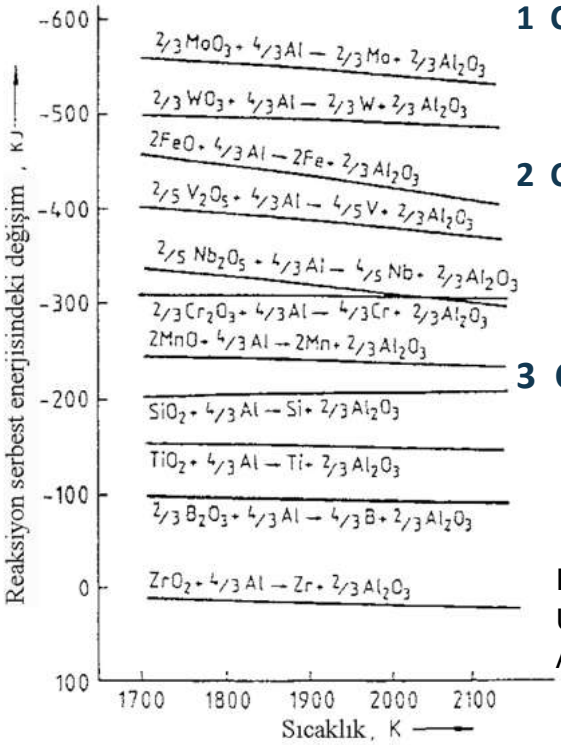
Values are per reaction as plotted (1 mol Si).
Si favors Mo and Mn reduction; direct reduction to pure Ti or Zr is unfavorable here.

ΔG does not tell us whether enough heat is released to melt the charge.

Principles of metallothermic reduction

Reading an aluminothermic reduction diagram

Use ΔG to judge feasibility; use a separate heat balance to assess furnace heating.



1 Choose the temperature

At 1900 K (≈ 1627 °C), locate each reaction curve and read its ΔG from the vertical axis.

2 Check the sign — this axis is reversed

Above zero: ΔG < 0 → favorable reduction.
Below zero: ΔG > 0 → unfavorable in the reference states represented by the graph.

3 Compare approximate readings at 1900 K

Cr_2O_3: −305 kJ TiO_2: −150 kJ

ZrO_2: +20 kJ → unfavorable here

Each curve uses 4/3 mol Al (1 mol O_2 transfer).
Unlike Si, Al can reduce TiO_2 in this diagram.
Actual alloy and slag activities still matter.

ΔG does not tell us whether enough heat is released to melt the charge.

Principles of metallothermic reduction

- Heat of Reaction per Mole of Reducing Agent

	$-\Delta H_{298}$, kJ
$^3/_4TiO_2 + Al \rightarrow {}^3/_4Ti + {}^1/_2Al_2O_3$	130.2
$^3/_4SiO^2 + Al \rightarrow {}^3/_4Si + {}^1/_2Al_2O_3$	156.0
$^1/_2B_2O_3 + Al \rightarrow B + {}^1/_2Al_2O_3$	202.6
$^3/_{10}Nb_2O_5 + Al \rightarrow B + {}^1/_2Al_2O_3$	268.8
$^1/_2Cr_2O_3 + Al \rightarrow Cr + {}^1/_2Al_2O_3$	273.8
$^3/_{10}V_2O_5 + Al \rightarrow {}^3/_5V + {}^1/_2Al_2O_3$	373.9
$^1/_2WO_3 + Al \rightarrow {}^1/_2W + {}^1/_2Al_2O_3$	417.4
$^3/_8Fe_3O_4 + Al \rightarrow {}^9/_8Fe + {}^1/_2Al_2O_3$	420.2
$^1/_2Fe_2O_3 + Al \rightarrow Fe + {}^1/_2Al_2O_3$	428.3
$^3/_2FeO + Al \rightarrow {}^3/_2Fe + {}^1/_2Al_2O_3$	442.1
$^1/_2MoO_3 + Al \rightarrow {}^1/_2Mo + {}^1/_2Al_2O_3$	466.4
$^2/_3WO_3 + Si \rightarrow {}^2/_3W + SiO_2$	348.8
$^1/_2Fe_3O_4 + Si \rightarrow {}^2/_3Fe + SiO_2$	352.1
$^2/_3Fe_2O_3 + Si \rightarrow {}^4/_3Fe + SiO_2$	363.0
$2FeO + Si \rightarrow 2Fe + SiO_2$	381.8
$^2/_3MoO_3 + Si \rightarrow {}^2/_3Mo + SiO_2$	414.1

The table reports $-\Delta H_{298}$: a positive number means heat is released at 298 K.

1 Read the sign correctly

For Cr_2O_3 + Al, the entry is +273.8.
Therefore $\Delta H_{298} = -273.8$ kJ per mol Al consumed.

2 Scale to the balanced reaction

$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$

$\Delta H_{298} = 2 \times (-273.8)$
$= -547.6$ kJ

3 Connect to the previous diagrams

ΔG: is reduction favorable?
ΔH: how much heat is released?
Furnace demand also depends on heating, melting and heat losses.

Principles of metallothermic reduction

Reaction heat: compare Al and Si on a molar basis

The table reports $-\Delta H_{298}$: a positive number means heat is released at 298 K.

Selected values from the supplied table

Oxide	Al	Si
Heat released, kJ per mol of reducing agent		
TiO_2	130.2	—
SiO_2	156.0	—
Cr_2O_3	273.8	—
WO_3	417.4	348.8
Fe_2O_3	428.3	363.0
FeO	442.1	381.8
MoO_3	466.4	414.1

— = no value listed in the supplied table.

A larger value means more heat per mol reductant; it does not directly compare heat per mol metal.

1 Read the sign correctly

For Cr_2O_3 + Al, the entry is +273.8.
Therefore $\Delta H_{298} = -273.8$ kJ per mol Al consumed.

2 Scale to the balanced reaction

$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$

$\Delta H_{298} = 2 \times (-273.8)$
$= -547.6$ kJ

3 Connect to the previous diagrams

ΔG: is reduction favorable?
ΔH: how much heat is released?
Furnace demand also depends on heating, melting and heat losses.

Exothermic does not automatically mean the charge can melt and sustain itself.

Principles of metallothermic reduction

Thermochemistry: aluminum's affinity for oxygen

Compare oxide formation enthalpies using the same amount of oxygen.

Normalize before comparing

For M_xO_y: divide ΔH°f by y.
The result is kJ per mol of oxygen atoms. More negative values indicate more heat released during oxide formation.

Selected solid oxides at 298.15 K

Oxide	ΔH°f / mol O (kJ)
Al_2O_3 (α)	−558.6
TiO_2 (rutile)	−472.4
SiO_2 (quartz)	−455.4
Cr_2O_3	−378.2
MoO_3	−248.4

1 mol O atoms = ½ mol O_2. Source: NIST–JANAF; rounded values for the stated phases.

Hess's law: calculate reaction heat

$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$

ΔH°rxn = ΣΔH°f(products) − ΣΔH°f(reactants)
Elements in their standard states: ΔH°f = 0.

ΔH°rxn = −1675.7 − (−1134.7)
= −541.0 kJ per reaction

The same result on an oxygen basis:

3 × [−558.6 − (−378.2)] ≈ −541 kJ
Three mol O atoms transfer to aluminum.
The negative sign means heat is released.

Use ΔG at the process temperature to judge feasibility; ΔH alone does not establish it.

Principles of metallothermic reduction

Thermochemistry: compare oxides per mole of oxygen

Positive $-\Delta H°f_{,298}$ values represent heat released when an oxide forms from its elements.

Oxide	$-\Delta H°f$ / mol oxide kJ/mol oxide	$-\Delta H°f$ / mol O kJ/mol oxygen atoms
CaO	634.7	634.7
MgO	601.7	601.7
Al_2O_3	**1678.5**	**559.5**
ZrO_2	1101.5	550.8
TiO_2	945.4	472.7
SiO_2	911.0	455.5
B_2O_3	1272.8	424.3
Na_2O	415.3	415.3
Ta_2O_5	2047.3	409.5
Nb_2O_5	1900.8	380.2
Cr_2O_3	1130.4	376.8
V_2O_5	1551.6	310.3
WO_3	843.2	281.1
Fe_3O_4	1117.5	279.4
Fe_2O_3	821.9	274.0
FeO	264.6	264.6
MoO_3	745.7	248.6

How to normalize

M_xO_Y: divide the oxide value by y.
Al_2O_3: 1678.5 ÷ 3 = 559.5
One mol O atoms = ½ mol O_2.

Why the common basis matters

Ta_2O_5 looks larger per mol oxide, but Al_2O_3 releases more heat per mol oxygen: 559.5 vs 409.5.

Apply Hess's law

$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$
$\Delta H°rxn = -1678.5 + 1130.4$
$= -548.1$ kJ

Use ΔG(T) to judge reduction feasibility; ΔH measures reaction heat.

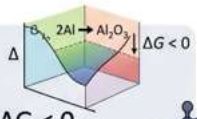
Source: supplied table at 298 K. Per-O values recalculated from its oxide values and rounded.

Principles of aluminothermic reduction:

Aluminothermic reduction: when is furnace heat needed?

Al transfers oxygen from a metal oxide to Al_2O_3, releasing heat and producing metal.

Thermodynamic driving force



Many reactions in Figure 2.3b have $\Delta G < 0$. Their products are favored in the reference states shown, but feasibility also depends on temperature and alloy/slag activities.

Heat balance determines the process



Available heat must cover:

- Heating the charge to the reaction temperature
- Melting the metal/alloy and slag
- Heat losses to the vessel and surroundings



Example: chromium production



$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$

From the preceding supplied thermochemistry table:
$\Delta H_{298} \approx -548$ kJ per reaction
≈ 183 kJ (43.7 kcal) released per mol O atoms.

Sufficient reaction heat



After ignition, a suitable charge can sustain reduction and melting without continuous furnace heating. Control heat loss and allow metal–slag separation.

Interpret the quoted heat range carefully



45–150 kcal/mol O ≈ 188–628 kJ/mol O.
Treat this as an approximate source range; heat release varies with the oxide and reaction.

Insufficient reaction heat



Preheat the charge and/or supply electric furnace heat to maintain smelting conditions.

Principles of metallothermic reduction

Titanium recovery: balance yield and alloy composition

Residual TiO in the slag lowers titanium recovery into the alloy.

Equilibrium between alloy and slag

$$3TiO(slag) + 2Al(alloy) \rightleftharpoons 3Ti(alloy) + Al_2O_3(slag)$$

$$K = (aAl_2O_3 \times aTi^3) / (aTiO^3 \times aAl^2)$$

At fixed temperature and alumina activity:

$$aTiO = (aAl_2O_3 / K)^{1/3} \times aTi / aAl^{2/3}$$

Lower Ti activity in the alloy → less TiO in slag.
Higher Al activity in the alloy → less TiO in slag.
Lower Al_2O_3 activity also favors reduction.

Activities represent effective concentrations; composition changes need not change them proportionally.

Illustrative equilibrium comparison

Hold T, K and aAl_2O_3 constant:

Halve aTi → aTiO falls to 50%.
Double aAl → aTiO falls to $2^{-2/3} \approx 63\%$.

The production trade-off

More Al can improve titanium recovery, but excess residual Al may exceed the required ferroalloy composition.

Reduction followed by refining

Use sufficient reductant to lower slag losses; then refine excess Al if needed, while controlling titanium losses during refining.

Principles of metallothermic reduction

Aluminothermic production: match the product to its use

Aluminum can reduce selected oxides to make metals and alloys for steel and superalloys.

Ferroalloys for steelmaking

FeB FeMo FeNb FeTi FeV FeW

These alloys deliver B, Mo, Nb, Ti, V or W to the steel melt. Iron is an intentional constituent of the ferroalloy product.

Example: ferroniobium

$$3Nb_2O_5 + 10Al \rightarrow 6Nb + 5Al_2O_3$$

With an Fe source in the charge, reduced Nb collects in an Fe–Nb alloy; Al_2O_3 enters the slag. The equation shows the oxygen-transfer step.

Some routes use Si and Al together; the reductant and charge depend on the product specification.

Feedstocks for superalloys

Iron-free products

Use a metal product when added Fe would exceed the alloy specification. High-purity chromium metal is a suitable example.

High-purity ferroalloys

Where Fe is acceptable, FeNb can be made from purified Nb_2O_5 to reduce impurities introduced by the mineral feed.

Feed purity is only the starting point

Control residual Al, O and other impurities through charge design, slag separation and any required refining.

Principles of metallothermic reduction

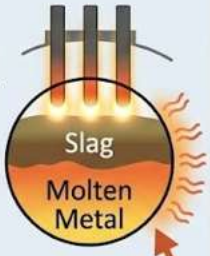
METALLOTHERMIC PRODUCTION
Flexible Routes for Small Batches

Combining chemical reduction & furnace heating to suit product and batch size.

WHEN AN ELECTRIC ARC FURNACE (EAF) IS USEFUL

FeB, FeNb, FeTi, FeV

Used for small tonnages, especially when a suitable furnace is already available.



WHAT THE FURNACE CONTRIBUTES

External heat maintains reduction, melting, and metal–slag separation when the charge does not supply enough reaction heat.

HEATING METHOD ≠ REDUCING AGENT

- Arc furnace supplies energy (heat)
- Charge chemistry determines whether reduction is **metallothermic**.

POTENTIAL ADVANTAGES OVER CARBON REDUCTION

RAPID REACTIONS

Shorter hot residence time can reduce total heat lost by radiation and convection.

LESS REACTION GAS

Oxygen enters a solid/liquid oxide slag rather than forming large amounts of CO.

FLEXIBLE BATCH PRODUCTION

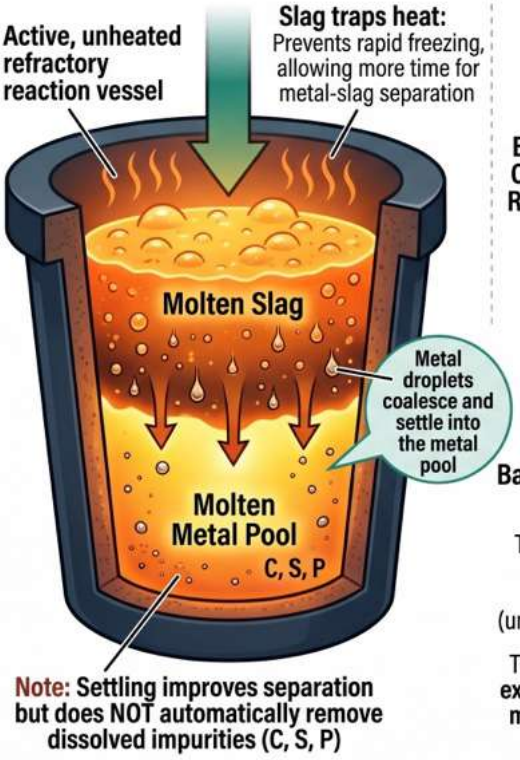
Charge design can be adapted for different products in suitable equipment.

LOWER INITIAL INVESTMENT AT SMALL SCALE

Simple batch vessels or an existing furnace can reduce the need for dedicated plant.

Metallothermic Production: Refining & Metal Recovery

1. Retain Slag & Manage Heat



Note: Settling improves separation but does NOT automatically remove dissolved impurities (C, S, P)

2. Control Quality Through Charge



Balanced Oxygen & Reductant

Too little Al/Si = **incomplete reduction** (unrecovered oxide)

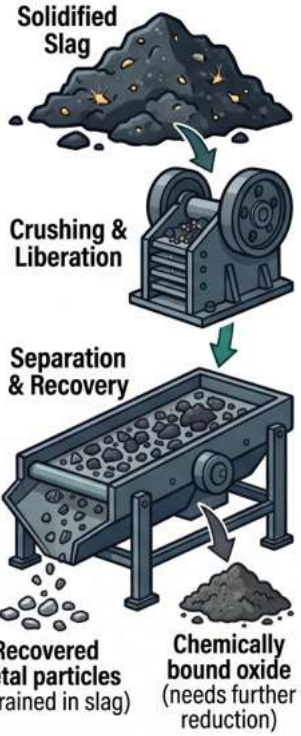
Too much Al/Si = **excessive residual metal** in the alloy

Limit Impurities

Controlled **Fe, Oxide, & Reduuctants**

Control **C, S, P** feeds; short batch times offer limited refining

3. Recover Metal from Solidified Slag



Principles of metallothermic reduction

Metallothermic production: refining and metal recovery

Short batch times leave limited opportunity to adjust or refine the metal after reduction.

Retain slag long enough for separation

Metal droplets form within the slag. While the phases remain sufficiently fluid, droplets can coalesce and settle into the metal pool.

Slag also helps retain heat

Removing the slag exposes the hot metal and increases heat loss. In an unheated batch, rapid freezing can shorten the time available for separation and refining.

Do not equate settling with full refining

Settling improves metal–slag separation; it does not automatically remove dissolved C, S, P or excess reductant from the alloy.

Control quality through the charge

Balance oxygen and residual reductant

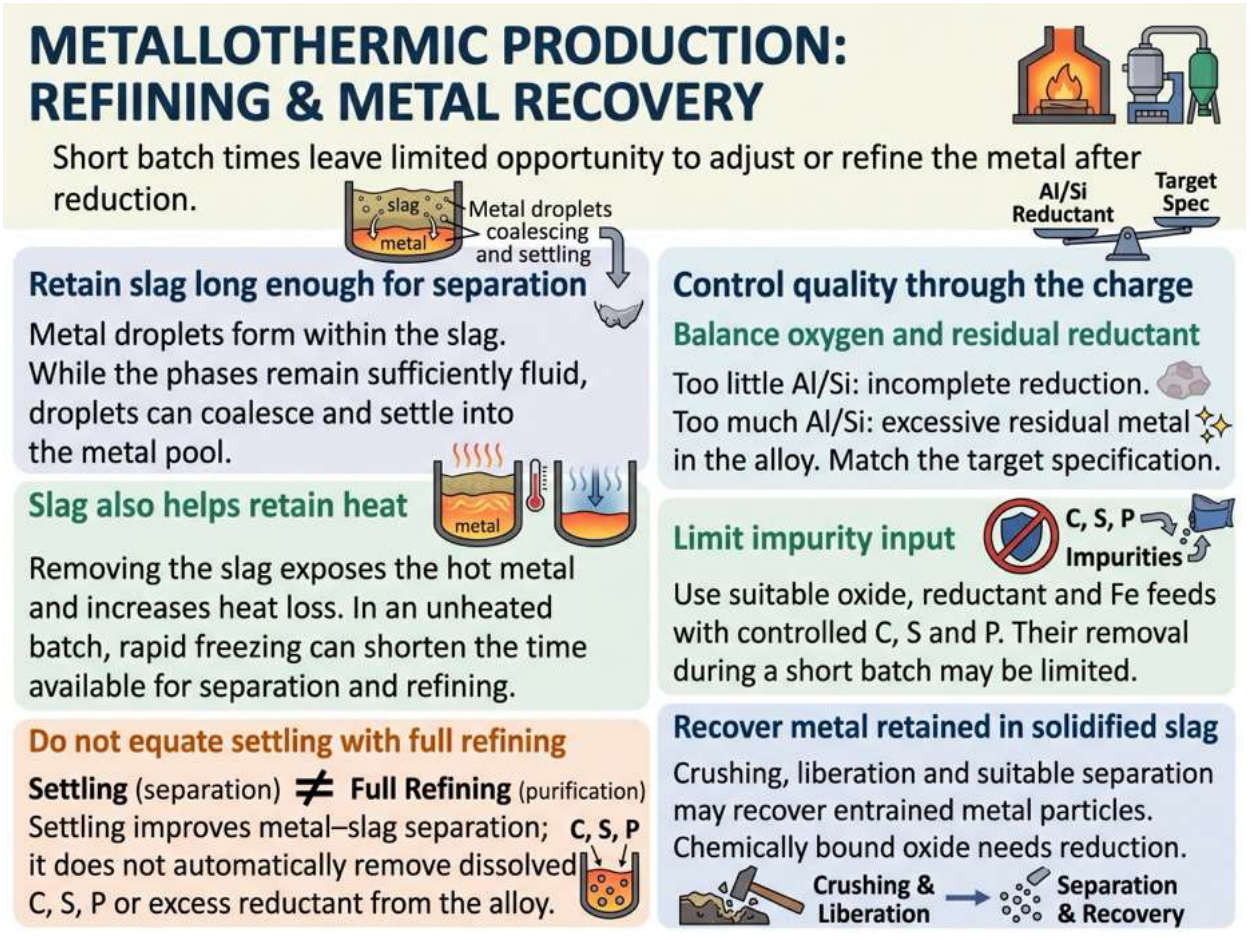
Too little Al/Si: incomplete reduction. Too much Al/Si: excessive residual metal in the alloy. Match the target specification.

Limit impurity input

Use suitable oxide, reductant and Fe feeds with controlled C, S and P. Their removal during a short batch may be limited.

Recover metal retained in solidified slag

Crushing, liberation and suitable separation may recover entrained metal particles. Chemically bound oxide needs reduction.

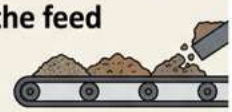


Principles of metallothermic reduction

Trace impurities: high-purity chromium as an example

Small impurity concentrations can determine whether a metal meets its end-use specification.

Impurities can enter with the feed



As Sn Pb Sb Cu

Ore and other charge materials may introduce these elements. Their transfer to the alloy depends on reduction and partitioning between metal, slag and gas.

Slag cover and nitrogen pickup

A continuous slag cover reduces direct contact between hot metal and the atmosphere, which can limit nitrogen pickup.



Protection is incomplete

Nitrogen still depends on exposure, temperature, alloy chemistry and nitrogen in the feed.

Demanding applications for chromium



Superalloys and electrical heating alloys may require chromium metal with tightly controlled trace impurities.

Ga, Pb and B: check the specified limit



Some high-purity grades restrict selected elements to only a few ppm. Limits vary with the grade and intended application.

How small is 5 ppm?



1 ppm by mass = 1 mg/kg

5 ppm in 1000 kg Cr = only 5 g impurity.

Illustrative value, not a universal limit. B may be added deliberately to some final alloy grades.

Principles of metallothermic reduction

Aluminothermic ferroalloys: output and contained metal

Read production and composition together

Product	Alloy output	Nominal composition	Contained element
	t/year	mass %	t/year, calculated
FeB	1,350	18% B	243 B
FeMo	**40,500**	**65% Mo**	**26,325 Mo**
FeNb	16,300	65% Nb	10,595 Nb
FeV	25,000	80% V	20,000 V
FeW	4,000	80% W	3,200 W

Worked example: FeMo

40,500 t alloy/year × 0.65 = 26,325 t Mo/year

Source note on steelmaking additions

The source attributes about ⅓ of Mo use to FeMo and ⅔ to MoO_3 additions. Treat this as historical; MoO_3 is reduced in the steelmaking process.

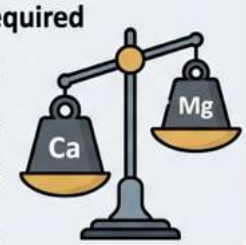
Principles of metallothermic reduction

Reductant selection: oxygen capacity and reaction heat

Strong oxygen affinity alone does not make a reductant economical or easy to operate.

Mass needed to bind 16 kg of oxygen

Reductant	Mass required
Ca → CaO	≈ 40 kg
Mg → MgO	≈ 24 kg
Al → Al_2O_3	**≈ 18 kg**
Si → SiO_2	**≈ 14 kg**



Why Ca and Mg are less attractive here

Their cost and consumption can outweigh their strong reducing power. Volatility also complicates high-temperature high-temperature smelting.



Normal boiling points:
Ca ≈ 1484 °C; Mg ≈ 1090 °C.



An empirical aluminothermic guideline

Heat released at 298 K > 300 kJ per mol Al

This source criterion helps screen charges for sufficient heat to keep metal and slag fluid for separation. It is not a guarantee.



Compare the preceding reaction-heat table

vs Comparison

Cr_2O_3 + Al: 273.8 kJ/mol Al
Below the guideline → extra heat may be needed.



V_2O_5 + Al: 373.9 kJ/mol Al
An **80% FeV** charge can be very vigorous; **heat removal** may be needed.



Warning

Do not transfer this threshold to Si reduction.
Always check the complete charge heat balance.

Principles of metallothermic reduction:

Auxiliary reactions: add heat and form a ferroalloy



Some primary reduction charges need extra heat to keep both metal and slag fully liquid.

An iron oxide–aluminum reaction



Fe_2O_3 + 2Al → 2Fe + Al_2O_3

Two useful effects





1 Additional reaction heat
Helps heat and melt the charge.

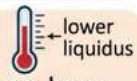
2 Iron joins the reduced metal (Fe → M)
Produces the intended Fe–M ferroalloy.

Example from the preceding source table



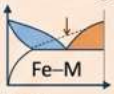
- 428.3 kJ released per mol Al
- 2 mol Al → ≈ 857 kJ per reaction above.
- This is a 298 K value, not a furnace heat balance.

Why alloy formation can help



An appropriate Fe–M composition can have a lower liquidus than the pure target metal, reducing the temperature needed for melting.

Check the phase diagram



The effect depends on the metal and alloy composition. Adding Fe does not always substantially lower the melting temperature.

Exception: iron-free chromium metal



Adding iron oxide would introduce unwanted Fe. Use a heat-supply strategy compatible with the required chromium purity.



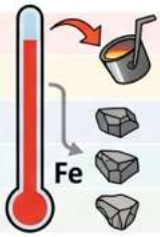
Principles of metallothermic reduction:

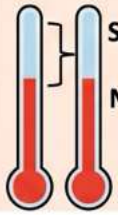
Alloy melting temperatures guide the charge heat balance

Adding Fe lowers the reported melting temperature, but some ferroalloys still need substantial heat.

Melting points of pure metal and ferroalloys

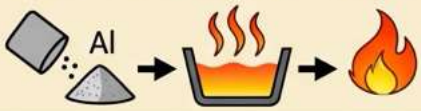
Metal	Pure metal	Iron alloy (mass %)	Reported melting temperature
V	1902 °C	80% V — FeV	1770 °C
B	2180 °C	16% B — FeB	1550 °C
Nb	2468 °C	65% Nb — FeNb	1560 °C
Mo	2620 °C	**65% Mo — FeMo**	**1900 °C**
W	3410 °C	**80% W — FeW**	**2500 °C**



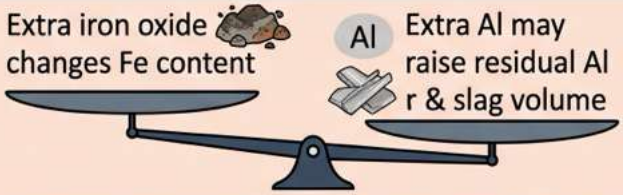


Why add Al to a silicothermic charge?

Al reduction can increase heat release. This helps melt high-temperature products such as FeMo and FeW, as well as the slag.



More heat must still meet the product grade



Compare total cost, recovery and specifications.

Source values retained. Alloys may melt over a range; use phase diagrams for the required liquidus.

Principles of metallothermic reduction

Auxiliary reactions: compare heat on the same Al basis

Additional oxidants can release heat when reduced by Al, helping a charge remain molten.

Balanced reactions from the supplied source	ΔH_{298} (kJ/mol Al)
$K_2Cr_2O_7 + 4Al \rightarrow 2Cr + 2Al_2O_3 + K_2O$	**−422.0**
$3BaO_2 + 2Al \rightarrow 3BaO + Al_2O_3$	**−812.0**
$NaClO_3 + 2Al \rightarrow Al_2O_3 + NaCl$	**−869.2**
$3KClO_4 + 8Al \rightarrow 4Al_2O_3 + 3KCl$	**−839.0**



Read the normalization correctly



1-mole Al container consumed



Negative ΔH = Heat Released

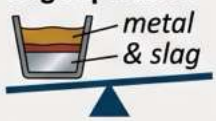
⚠ ✓ Values are per **mol Al** consumed
✗ **NOT** per whole reaction as written
✗ **NOT** per mol oxidant

Heat is only one selection criterion

Purity



Slag Separation



Charge Heat Balance



Additions change slag and reaction products. Check alloy purity, slag separation, and the complete charge heat balance.

Source values retained at 298 K. Ba-containing products are written on a balanced oxide basis.

Principles of metallothermic reduction:

Auxiliary heat: preheating and nitrate limitations

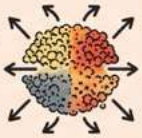


A heat-supply method must preserve charge control and the required product purity.

Preheating a reactive mixed charge

Purpose: add sensible heat before reduction to reduce the heat deficit during smelting.

Why it can be difficult



Low effective thermal conductivity can cause slow, uneven heating.



Premature reaction could cause charge loss damaged preheating equipment.

Miniatiure aniniator

Scope of the source statement



The reported abandonment concerns reactive mixture preheating in these processes; it does not mean all feed preheating is obsolete.

Nitrates as auxiliary oxidants



Low reagent cost is not enough

Nitrogen-bearing additions may introduce nitrogen into products that require low nitrogen concentrations.

Nitride-forming metals are sensitive



Titanium (**Ti**), **Niobium (Nb)** & **Vanadium (V)** readily form nitrides. Nitrogen pickup alter composition and increase nitride inclusions.

Choose heat supply around the specification

Assess nitrogen input, final impurity limits and the complete heat balance when selecting an auxiliary heat source.



THE ROLE OF NITROGEN-BEARING ADDITIONS

Evaluating Nitrates (e.g., $NaNO_3$, KNO_3) as Auxiliary Oxidants in Metallurgy

The Purpose: Heat Generation & Reduction

Auxiliary Heat & Ignition

Nitrates
$NaNO_3$ / KNO_3



Decomposition and rapid reaction with Al releases **intense auxiliary heat** to **keep metal & slag fluid**.

Thermal Boost



Decomposition and rapid reaction with Al **releases intense auxiliary heat** to **keep metal & slag fluid**.

Oxygen Delivery

NO_3^- → O → N_2 / NO_x

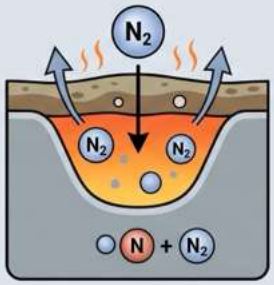
Al → Al_2O_3 and other Aluminium and other metals

Provides **active oxygen** for reduction

The Risk: Nitrogen Contamination & Nitride Formation

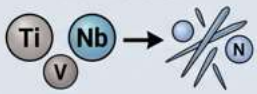
Nitrogen Pickup & Quality Issues

Gas Dissolution



Nitrogen dissolves easily into the hot, unprotected alloy.

Nitride-Forming Metals are Sensitive



Ti, Nb, V + N → Nitride Inclusions
(Imperfect Material)



Alloy Purity Altered / Increased Brittleness & Defect Risk

Key Takeaway

Low cost cannot outweigh the risk of contamination. Always check the **final nitrogen limits** for sensitive applications like superalloys.

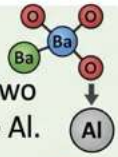


Principles of metallothermic reduction:

Auxiliary oxidants: oxygen capacity and product compatibility

Choose an addition for its reaction products, handling characteristics and total process cost.

BaO_2: limited oxygen delivery



BaO_2 is reduced to BaO; one of its two oxygen atoms is available to oxidize Al.

Molar mass ≈ 169.3 g/mol
Available oxygen = 16.0 g/mol
Oxygen capacity ≈ 9.5% by mass



Lower Alumina Activity

BaO can combine with Al_2O_3 in the slag, which can lower alumina activity and support the reduction reaction.



Source examples: Ti–Al and Zr–Al products



Specialized, high-heat processes

Warning: Low economic appeal & high slag burden specialized use with highly exothermic charges.

$NaClO_3$ and $KClO_4$



The source favors $KClO_4$ for greater stability and cites moisture uptake by $NaClO_3$ as a handling disadvantage.

Greater Stability

Handling Disadvantage: Moisture uptake

Stability does not establish compatibility

Stability does not establish compatibility



Assess behavior in the actual mixed charge, moisture control, products and impurity limits.

$K_2Cr_2O_7$ in chromium charges



The source reports preference over CrO_3 for cost and handling considerations. Both contain Cr(VI); this does not make $K_2Cr_2O_7$, a benign substitute.



Not a benign substitute

Reported source preferences are process-specific.

Principles of metallothermic reduction:

Fluxes: keep slag fluid and support metal recovery

A suitable slag must remain liquid long enough for metal droplets to separate and settle.

Pure reaction oxides are refractory



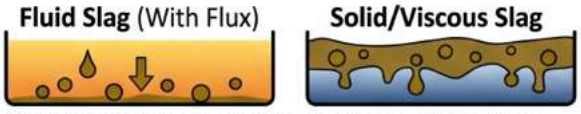
Al_2O_3: ≈ 2050 °C
SiO_2: ≈ 1722 °C

Approximate pure-oxide melting temperatures from the source—not the melting temperatures of a multicomponent process slag.

Mixing oxides can create a useful slag



Al_2O_3, SiO_2 and suitable residual iron oxide can form lower-liquidus slag compositions. CaO is added to adjust slag chemistry.



Fluid slag helps droplets coalesce and settle; solid or viscous slag can retain valuable metal.

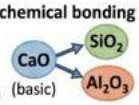
CaO has two connected roles

1. Adjust melting and fluidity



An appropriate CaO addition can lower the slag liquidus and improve separation. The effect depends on the complete composition.

2. Lower product-oxide activities



Basic CaO can bind SiO_2 and Al_2O_3 into silicate/aluminate species, lowering their activities in suitable slag compositions.



Lower $aSiO_2$ or aAl_2O_3 can favor reduction by Si or Al because these oxides are products.

Warning ⚠ More CaO is not always better: excessive additions can raise the liquidus, solids content and heat demand.



Principles of metallothermic reduction

Fluxes: useful slag constituents and the chromium exception

Slag design depends on oxide combinations, volatility and interaction with the refractory.

Residual oxides can assist fluxing

$Al_2O_3 \cdot TiO_2 = Al_2TiO_5$



The source reports melting at ≈ 1590 °C. This illustrates how oxide combinations can melt below pure Al_2O_3 (≈ 2050 °C).

Na_2O and K_2O



Retained alkali components can aid fluxing. Volatilization reduces their contribution and changes slag composition during reaction.

CaO in alumina-rich slags

Calcium Oxide + Al_2O_3 → Calcium aluminate

Lower melting point (liquidus)

Calcium aluminate formation can lower the liquidus in suitable compositions. Its effect is composition-dependent.

Chromium: flux compatibility matters

The source avoids lime in its Cr-metal process

⚠ Lime/⃠/CaO → Chromium reduction

CaO entering from flux or refractory can alter the slag and interfere with the intended chromium recovery and separation.



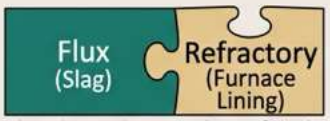
Chromate formation depends on conditions

Oxidizing Conditions

CaO + oxidants → stabilizes $CaCrO_4$ (Calcium chromate)

Do not assume it forms in every reducing slag.

Choose the flux and refractory together



slag liquidus and perfectly.

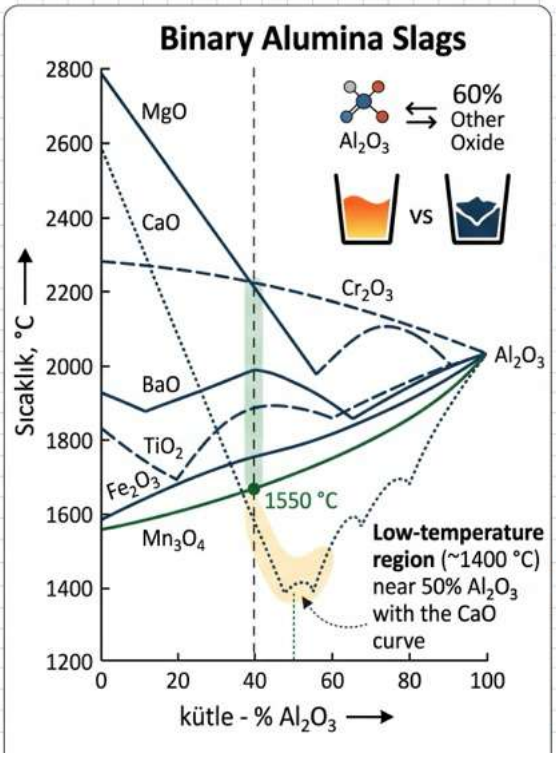
- ☑ Phase stability
- ☑ Oxygen potential
- ☑ Slag liquidus
- ☑ Product purity

Principles of metallothermic reduction:

Reading melting curves for binary alumina slags



Flux effectiveness depends on both the oxide added and the resulting slag composition.



1 Read the composition correctly

x-axis: mass % Al_2O_3; y-axis: temperature (°C). Each curve is Al_2O_3 + the labelled oxide. 40% Al_2O_3 means 60% of the other oxide.

2 Compare at the same Al_2O_3 content

At 40 mass % Al_2O_3, approximate readings:

✓ **CaO mixture: ≈ 1550 °C** 🔥 fluid

❗ **MgO and Cr_2O_3 mixtures: ≈ 2200 °C** ❗

❄ refractory / not melting

At 1800 °C, the CaO mixture is above its curve; the MgO and Cr_2O_3 mixtures are below theirs.

3 Look for low-temperature regions



The CaO curve has a minimum near 50% Al_2O_3 and ≈ 1400 °C. Flux additions are not monotonic: choose a composition, not simply "more flux."

Note: Sıcaklık = Temperature; kütle = mass %.

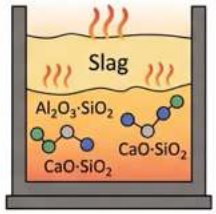
Principles of metallothermic reduction:

Compound formation can contribute additional reaction heat

Positive $-\Delta H_{298}$ values indicate heat release; keep the molar basis consistent.

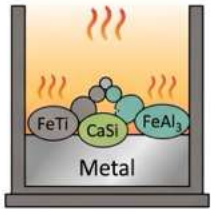
Compound oxides in the slag

Compound	$-\Delta H_{298}$ (kJ/mol)
$Al_2O_3 \cdot SiO_2$	192.6
$2Al_2O_3 \cdot B_2O_3$	69.5
$BaO \cdot Al_2O_3$	100.5
$CaO \cdot Al_2O_3$	15.1
$CaO \cdot SiO_2$	89.2



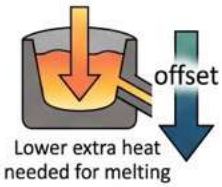
Intermetallic compounds in the metal

Compound	$-\Delta H_{298}$ (kJ/mol)
FeTi	40.6
FeB	29.3
$FeAl_3$	112.2
FeSi	77.0
TiAl	80.8
CaSi	150.7
AlB	67.0





Why compound oxides matter



Their formation can offset part of the heat needed to warm and melt the flux/slag. Actual phase formation depends on composition.

Include each heat contribution once

Use Hess's law and the actual final phases; do not add formation heat twice.



Values transcribed from the source; reference reactions/phases must be checked before calculation.

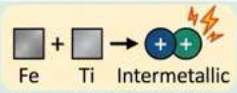
Principles of metallothermic reduction

Formation of Intermetallic Compounds

Additional heat must remain compatible with the final alloy composition

Heat benefit



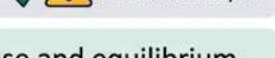


Formation of an intermetallic compound can release additional heat when that phase forms in the charge.

Composition constraint



Retaining Al or Si may increase heat release, but residual reducing elements must meet the product specification.



Teaching point: heat release and equilibrium oxygen content require separate checks.



- ☑ Heat Release Check
- ☑ Oxygen Equilibrium Check



Where aluminium compounds matter

The source identifies FeTi as the main case, with a smaller role in FeB production.

Oxygen and residual reductant



The source reports that many alloys can retain about 1% Al or Si while achieving low oxygen contents.



FeTi requires special attention

The cited 4–6% Al threshold depends on the equilibrium conditions. The source does not define those conditions.



Source percentages are retained as stated. Confirm their basis and the applicable alloy grade.

Principles of metallothermic reduction

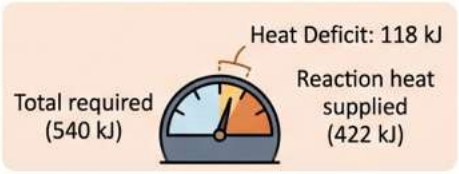
Thermochemistry in Practice: Chromium Reduction

A favourable reaction can still release too little heat to reach the target temperature

$$Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$$



$\Delta H_{298} = -422.0$ kJ per mole of reaction as written



Heat needed at 2000°C (2273 K)

Product	$H_{2998} - H_{2273}$
Al_2O_3	356 kJ
2Cr	184 kJ
Total heat requirement	**540 kJ**



Massive 2000°C (2273 K)

Basis: reactants at 298 K, one mole Cr_2O_3 reduced. target does not guarantee liquid slag.

Heat deficit before losses

540 – 422 = 118 kJ

Heat Deficit: 118 kJ

Deficit and Losses

The reaction alone cannot supply the listed heat requirement. Real heat losses increase the additional energy needed.



Charge Design & Plant Trials

Charge design uses thermochemical data. Plant trials refine heat-loss estimates and the required heat supplementation.



Source values retained. A 2000°C target does not guarantee liquid slag.

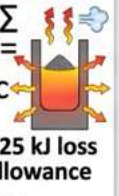
Principles of metallothermic reduction:

Thermochemistry in Practice: Heat Losses

The calculated heat balance provides a starting point for practical charge design.

Chromium example

deficit & loss



Initial deficit before losses: 118 kJ
\+ Loss allowance: 125 kJ (specific for the reference charge size)

If 125 kJ is a separate loss allowance:

Additional heat = 118 + 125 = 243 kJ on the same reaction basis

Charge size changes heat losses

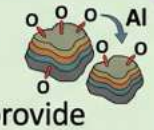


Smaller charges generally lose more heat per unit mass because they have more exposed surface relative to their volume.

Larger charges generally need less heat supplementation per unit mass.



Ore chemistry also matters



Higher oxidation states often provide more reducible oxygen and greater heat release with Al or Si. Compare balanced reactions on a consistent molar basis.

Auxiliary Al reactions with $K_2Cr_2O_7$



$K_2Cr_2O_7$ and/or $KClO_4$ supply extra heat. Their contribution enters the full balance.

⚠ The 125 kJ value is specific to the source example. Confirm itbasis before adding it to the 118 kJ deficit.

Principles of metallothermic reduction

Thermochemistry in Practice: Preoxidation

Manganese ore example

$3MnO + 2Al \rightarrow 3Mn + Al_2O_3$

The source reports insufficient heat for the intended process.



Insufficient Heat

$3Mn_3O_4 + 8Al \rightarrow 9Mn + 4Al_2O_3$

Preoxidation $MnO \xrightarrow{O_2} Mn_3O_4$



Adequate Heat

Preoxidation to Mn_3O_4 increases the oxygen removed per mole of Mole of Mn and the Al oxidized. The source reports adequate heat.



Also benefits ilmenite ($FeTiO_3$) depending on resulting phases.

Reaction rate and temperature

Too slow

More heat escapes



More heat escapes during reaction, which can hinder melting and separation.

Too fast or

Excessive peak temperatures



Heat can accumulate faster than it escapes, causing excessive peak temperatures.

Practical control



Heat Retention
Overheating



Balance heat retention against overheating. Trials identify the factors worth controlling for reliable, economical production.

Balance & Calibration

Rate changes heat release per unit time. Total reaction heat follows chemistry. Sufficiency depends on the full heat balance.

Principles of metallothermic reduction

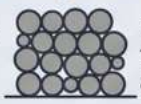
Particle Size and Charge Feeding

Particle size controls contact and reaction rate. Feeding controls heat release over time.

Batch charge: contact and packing



Increasing contact



Reducing trapped air

A suitable distribution of particle sizes helps the charge pack closely, increasing contact and reducing trapped air.
The source describes this as a "cement-like" distribution. It provides no universal numerical size range.

Specified aluminium powder grades

Al Powder
Grade A, B, C



Supplier size ranges support reproducible charge preparation and reaction behaviour.

Particle-size tradeoff

Heat Loss

Too coarse



Less contact can slow the reaction and increase the time available for heat loss.

Too fine

Dust carryover / Material loss



Uncontrolled rapid reaction

Dust carryover can reduce material recovery. Rapid reaction can make heat release harder to control.

Controlled charge feeding

Metered addition often **slowly and steadily feeding**



Metered addition often provides an effective way to regulate the rate of heat release.

The aim is adequate reaction rate with low dust loss. Select the distribution and feed rate for the actual process.

Principles of metallothermic reduction

Feed Form and Charge Consistency

Economical feed materials must still provide predictable reaction behaviour.

Aluminum lumps and turnings



Highly exothermic FeV charges can tolerate coarser aluminium feed. Lumps and turnings also serve some economical FeTi routes.

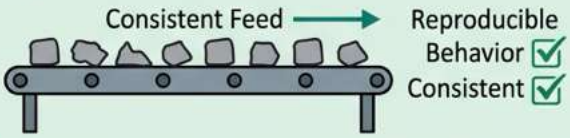
The practical tradeoff



Unprocessed feed

Unprocessed feed varies in size and form. Changes in its proportions can alter contact and produce variations between charges.

Consistent feed proportions help maintain reproducible charge behaviour.



Ore preparation



Coarse ore

In the source process, coarser ore requires higher temperatures to achieve an adequate reaction rate.

Very fine ore



Dust losses can reduce recovery. Pelletizing may be more economical when it limits excessive dust carryover.

Preparation follows the available ore form and the requirements of the process.

Raw Ore → Preparation Step → Metally Process

Feed-form choices depend on the alloy route and charge heat balance. Higher temperature also increases heat demand.